

DIGITAL BANKING AND THE FUTURE OF EMBEDDED FINANCE:

HOW WILL AI-POWERED FRAUD DETECTION AND ALGORITHMIC CREDIT SCORING IN CROSS-BORDER RAILS RESHAPE SYSTEMIC RISK?

Luv Aggarwal, Modern School Barakhamba Road, Luvagg06@gmail.com

Tanay Saravanan, The Kings School Grantham, tanaynoble@gmail.com

Alina Loana, Theoretical high school “Mihai Eminescu” Romania, adaalinaioana@gmail.com

Alisa, IMS Private School, Limassol, Cyprus, alisa29f@gmail.com

Shreshta, Irvington High School, srejaq268@gmail.com

Abstract

The digitization of global finance has transitioned from a phase of user interface improvement to a fundamental restructuring of payment rails and credit assessment architectures. This report investigates the convergence of embedded finance, real-time cross-border payments, and artificial intelligence (AI). Specifically, it examines how Graph Neural Networks (GNNs) in fraud detection and algorithmic cash-flow underwriting in credit scoring are enhancing financial inclusion and operational efficiency. However, this technological shift introduces new vectors of systemic risk, including model homogeneity, algorithmic herding behavior, and the opacity of cross-border data flows. By analyzing initiatives such as the BIS Innovation Hub’s Project Aurora and Project Nexus, alongside commercial applications by entities like Tala, Nubank and Nova Credit, this report argues that while AI is the only viable mechanism to secure high-velocity payment rails, it necessitates a new macro-prudential framework to manage the risks of synchronized algorithmic failure.

Introduction

The global financial system is undergoing its most significant redesign since the rise of electronic banking. What began as simple digitization has evolved into a deep restructuring of payment rails, liquidity pathways, and credit assessment models. This transformation is driven by the convergence of embedded finance, interoperable cross-border networks, and increasingly intelligent computational systems. Financial services are no longer confined to banks but are now integrated into platforms used for shopping, transportation, content creation, and decentralized digital interactions. As these ecosystems merge, long-standing assumptions about identity, risk, and regulatory oversight are being fundamentally reconsidered.

Artificial intelligence sits at the Centre of this shift. With transactions settling in milliseconds and data volumes growing exponentially, human-centered risk controls are no longer sufficient. Machine learning models- ranging from graph-based anomaly detectors to behavioral scoring engines- now perform core functions such as fraud prevention, credit underwriting, and real-time payment verification. AI has effectively moved from the periphery of financial operations to the backbone of global financial integrity.

Yet this rapid adoption introduces structural contradictions. The same models that enhance precision and expand access to credit also create tightly linked algorithmic environments where failures can cascade quickly and invisibly. Cross-border data dependencies, institution-wide model similarity, and reliance on a small group of AI and cloud providers raise pressing macro-prudential concerns. What happens when multiple institutions rely on the same flawed model? How do regulators oversee intelligence that operates across jurisdictions and at machine speed?

This paper explores AI's dual role as both a stabilizing force and a new source of systemic vulnerability. By analyzing international regulatory guidance, cross-market pilots, and emerging high-velocity payment systems, it argues that AI is not simply another technological upgrade but a foundational redesign of global financial infrastructure. Managing the risks of this transformation will require governance frameworks capable of addressing algorithmic interdependence and preventing synchronized failures in an increasingly automated financial world.

1. Digital banking

The rise in digital banking and its integration into daily financial activities is fundamentally transforming the delivery of banking services by enhancing efficiency, accessibility, and overall customer experience.

1.1 Definition of Digital Banking

During the last three decades, the financial services industry experienced various advancements, one of them being the emergence of digital banking. Digital banking refers to the digitization of banking services that previously required a branch visit. It can also be defined as the use of online and mobile channels to provide banking services, including money transfers, bill payments and account management.

1.2 Emergence and Adoption

The modern era of online banking began in 1994 with Stanford Federal Credit Union offering the first internet-based platform. It was followed by Wells Fargo and Presidential Bank in 1995. Despite the limited amount of services available online back then, the solutions presented to the broader audience became a basis for the current digital banking that is being used by hundreds of millions of people daily. Later, with the emergence of phones in the late 2000s, mobile banking started developing.

The COVID-19 pandemic has significantly boosted digital banking adoption, since consumers needed to find ways to perform various financial operations while being locked at home. As of 2025, approximately 3.6 billion people worldwide use online banking (CoinLaw, 2025),

underscoring the extent to which digital banking has become an integral component of daily financial activities, substantially enhancing the efficiency and accessibility of banking operations.

1.3 How Digital Banking Operates

Digital banking is based upon the combination of digital platforms, advanced software, and secure network infrastructures that provide customers with the possibility to manage their finances without visiting a physical branch.

- **Web platforms and mobile applications** conduct transactions and account management, offering instant transfers and payments along with real-time access to balances and other banking services.
- **Artificial Intelligence and machine learning** is employed with the aim to enhance personalisation, detect fraud, and streamline operations.
- **Cloud computing** assists in data storage, ensuring reliability and scalability of the system.

These technologies are integrated to improve speed, efficiency, security, and accessibility of banking services, forming a foundation of digital banking operations.

1.4 Key Advantages and Improvements over Traditional Banking

Digital banking has offered valuable benefits compared to traditional banking, that was previously in use. The main advantage presented was the possibility to perform various banking operations from anywhere in the world (given there is stable Internet connection) and at any time. Moreover, digital banking reduced costs by requiring fewer employees and limiting its reliance on physical branches. Digital solutions have facilitated automatic payments that can be performed on behalf of customers without their direct involvement. This has significantly simplified routine transactions, since consumers can delegate part of their responsibilities and focus on more meaningful tasks. Consequently, automatic payments not only have reduced the likelihood of missed deadlines or penalties, but also enhanced overall financial convenience. Additionally,

digital banking is more environmentally sustainable, as it minimises paper usage through digitisation of processes and operations, therefore reducing the financial industry's contribution to global deforestation. Finally, it facilitates faster cross-border payments, enabling efficient currency conversions and batch payments. Thus, many complexities, inefficiencies and high costs associated with international transactions have been significantly reduced after the emergence of digital banking.

1.5 Security Measures in Digital Banking

Security has become the major concern in mobile banking applications and online services. Digital banking now employs multi-factor authentication, biometric verification, and continuous AI-powered monitoring to mitigate crimes and other illegal activities within the financial industry. All transactions are constantly screened for potentially suspicious activity, and those appearing to be fraudulent can be blocked in order to enhance the overall security of the system and assist in preventing unauthorised access or financial losses. In addition, tokenisation is used to further improve security. This technology replaces sensitive information, such as card numbers, with unique tokens so that the actual data remains hidden during transactions. Overall, security is considered a top priority in digital banking, and various technologies are employed to ensure system stability and reliability.

1.6 Impact on the Banking Industry and Competition

The emergence of digital banking has had a substantial impact on the whole financial industry. Firstly, it has put considerable pressure on traditional banks by offering better terms and greater variety of services to customers. As a result, many traditional banks have been forced to develop their own mobile applications and websites to stay competitive. Also, digital banking has resulted in the rise of fintech startups and neobanks (which are digital-only banks) across the globe. Secondly, the level of competition has increased, since barriers to entry have become significantly lower, making the financial industry more attractive for new entrants. Thus, consumers have enjoyed a greater variety of suppliers and better deals offered by banks to surpass their competitors. Overall, the financial industry has expanded, reaching previously unserved populations due to remote locations and lack of physical branches in developing countries. Banks have reshaped their major focuses moving towards digitalisation and client-oriented solutions.

Case Study: Nubank and the "Challenger" Model The disruption caused by digital banking is best exemplified by **Nubank** in Latin America. Launched in Brazil in 2013 to combat the high fees and bureaucracy of the country's oligopolistic banking sector, Nubank operates with zero physical branches. By utilizing a cloud-native infrastructure (powered by AWS) and a proprietary real-time decision engine, Nubank reduced its operational costs by approximately 85% compared to traditional incumbents. This efficiency allowed them to offer fee-free credit cards and high-yield savings accounts to millions of previously unbanked Brazilians. As of 2024, Nubank serves over **100 million customers** across Brazil, Mexico, and Colombia. This case demonstrates that digital banking is not merely about convenience for the wealthy; it is a scalability engine that allows financial institutions to service low-income demographics profitably—something traditional brick-and-mortar banks historically failed to do.

1.7 Consumer Adoption Trends

Initially, consumer adoption of digital banking was slow, as many people were sceptical about security and privacy, and did not consider this form of banking fully reliable. However, over time, consumers' attitudes started changing, driven by greater access to information and growing experience with digital banking worldwide. Technological advancements in the field have also led to acceleration in consumer adoption. The introduction of more intuitive interfaces in mobile applications and web pages, faster transaction processing, and continuous access to banking services have enhanced the convenience and usability of digital platforms. Trust in digital channels has substantially increased with the implementation of stronger authentication methods, such as biometrics, and real-time fraud monitoring. The COVID-19 pandemic has compelled customers to explore digital banking due to restricted access to physical branches. As a result, digital banking has evolved from an alternative option to a fundamental expectation in modern financial experiences.

1.8 Future of Digital Banking

According to various studies conducted and real-world trends, digital banking is expected to continue developing, expanding its capabilities and service offerings, attracting more customers and increasingly relying on large-scale data. Generally, banking services are anticipated to become

invisible for customers through complete integration into their daily lives. Increased possibilities for personalised experience will emerge as banks enhance their analytical capabilities to provide tailored recommendations and context-aware services. Growing interconnection between digital ecosystems will shift banks towards adaptive and autonomous service models, offering solutions adjusted to individual preferences and changing life circumstances. Overall, over the next few decades, digital banking is likely to advance significantly, forming a foundation for future innovations within the financial industry.

2. AI in Fraud detection, Algorithmic Credit Scoring and Financial Inclusion

The integration of AI and data analytics in Fraud detection, Credit scoring and their collective role in Financial Inclusion is fundamentally transforming the finance sector in the world’s economy

2.1 AI in Fraud detection, and Financial integrity

The fraudulent activities in financial transaction processes are one of the biggest threats that affect business security; the economic losses are enormous, and trust in the financial sector is at its lowest. The situation has been compounded by the shift in operation of business to online basis given the fact that the number and size of financial transactions have increased significantly. It also presents new challenges which are associated with increased difficulty of identification and prevention of fraud related activities.

AI, Machine Learning (ML), and blockchain can be recognised as the best solution to combat these problems. AI and Machine Learning (ML) serve as critical mechanisms for processing high-velocity datasets, identifying complex fraud typologies, and adapting to adversarial strategies in real-time. Unlike static rule-based systems, the adaptive nature of ML—characterized by high-dimensional data analysis and low-latency processing—provides a necessary layer of transparency and accountability, significantly mitigating illicit activities within the financial sector.

Aspect	Traditional Fraud Detection	AI-Based Fraud Detection
1. Detection Method	Rule-based (fixed IF–THEN	Machine learning models that

	rules)	learn from data
2. Flexibility	Rigid; needs manual rule updates	Adaptive; models evolve with new fraud patterns
3. Speed	Often batch-based, delayed detection	Real-time or near real-time monitoring
4. Accuracy	High false positives; limited pattern recognition	Higher accuracy; detects complex and emerging patterns
5. Data Usage	Uses basic transactional data only	Uses behavioral, device, network, and multi-dimensional data

Table 1

As illustrated in Table 1, the limitations of rule-based systems necessitated an AI based fraud detection system, the most important being the detection method.

The traditional if-then rule was heavily criticised because of three main reasons:

- Could not detect new fraud patterns.
- High false positives (genuine customers flagged).
- The accumulation of excessive rules created system latency and operational inefficiency.

But with the introduction of AI and ML models it can learn from its own mistakes and improve in real time.

Thus, it's safe to say that AI and ML with its adaptive nature, high accuracy, use of three dimensional data, and real time monitoring speed can lead to better transparency, make the system accountable, and lead to a significant mitigation to corruption and illicit activities in the financial sector.

2.2 Algorithmic Credit Scoring

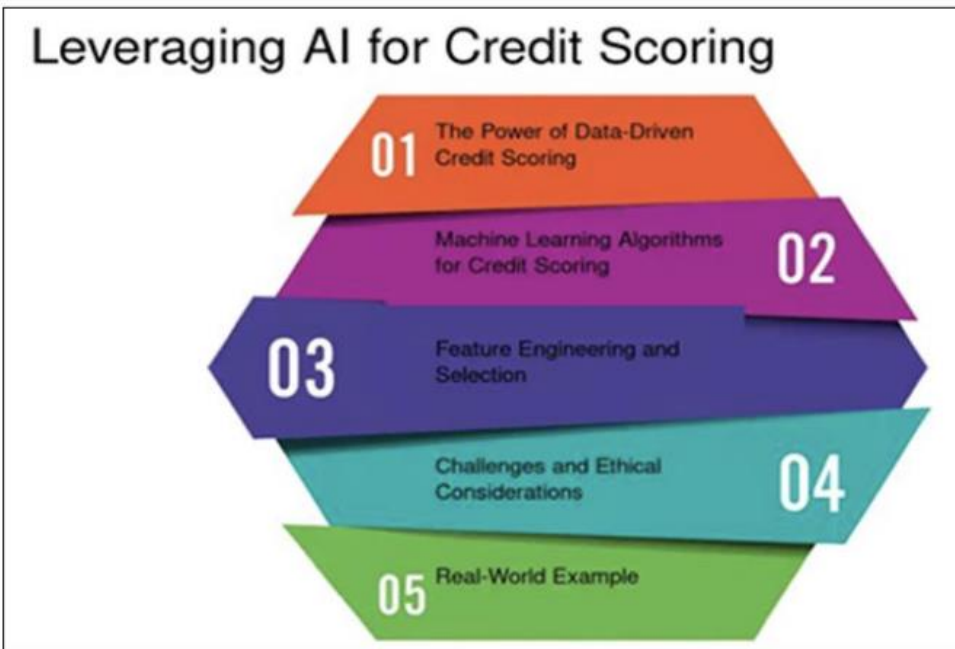
Traditional credit scoring techniques, which rely on credit history and conventional metrics of financial capabilities, exclude a large part of the population from emerging economies. This exclusion is partly because the pool of borrowers and smaller businesses do not have credit histories, which are imperative to mainstream credit scoring methods.

On this account, many such consumers within these markets are classified broadly as "unbanked" or "underbanked" and, therefore, unable to tap more appropriate and conventional channels of credit and financial services. This exclusion is socially and economically detrimental for an individual to create economic opportunities and for an economy and financial sector to develop.

That is why the application of Artificial Intelligence (AI) can be considered the solution to this challenge. AI thus has capabilities of better and broader credit risk assessment through analysis of other forms of data. Other data about mobile phone utilization, social networking sites, utility bills, etc., which are not financial in this particularistic sense, can be utilized as proxies to get a decent dependability on the person's creditworthiness. These complex phenomena reflect important differences in the business environments and consumer finance. New machine learning methods can process them to build more effective credit scoring systems for emerging economies.

Case Study: Tala and Mobile Data Scoring A prominent example of this innovation is Tala, a fintech company operating in emerging markets such as Kenya, the Philippines, India, and Mexico. Tala addresses the "credit invisible" population by utilizing an Android app that analyzes behavioral data from the user's smartphone with their permission. Instead of looking at a bank history, Tala's AI algorithms examine over 10,000 data points, such as the regularity of calls to family members (which correlates with social stability), the consistency of utility bill payments

via SMS, and even the daily routes taken by the user. By processing these alternative metrics, Tala has successfully dispensed over \$4 billion in micro-loans to millions of customers who previously had no formal credit score, proving that behavioral consistency can effectively predict repayment reliability



Picture 1

Advantages of Algorithmic Credit Scoring:

1. This system can provide a vast segment of people with an opportunity to receive credit and increase credit possibilities for those who cannot prove their solvency by usual methods. This inclusiveness can help many small players to engage more robustly and contribute more effectively to the markets and, hence, growth.
2. Credit assessments will have higher accuracy through more advanced AI models, thus lowering default risks and the general efficiency of lending. In turn, it can build more stability and increase the profitability of lending activities for financial organizations.

However, some issues arise from adopting AI in the credit scoring system. The first one is As every algorithm has its data sources, it can strengthen existing prejudice that should be eliminated, or it can trigger new prejudice in AI, which is unethical. This importance must assume a sound and flexible legal infrastructure to regulate the applicability of AI in the financial industry whilst safeguarding consumers' interests and personal information. Moreover, the input and output assets must be optimized for the AI tools, which may be costly in terms of IT infrastructure in resource-scarce environments.

Ultimately, the advantages of algorithmic credit scoring outweigh the disadvantages. Thus, AI-driven scoring is an essential evolution in an era where financial fraud is rampant and the underbanked population remains vast.

2.3 AI and ML as tools for financial inclusion

Beyond credit scoring, AI serves as a multifaceted engine for broader financial inclusion. Two key technologies driving this change are Machine Learning (ML) and Natural Language Processing (NLP).

- Machine Learning reduces the "cost to serve" for banks. By automating risk assessment and back-office operations, ML makes it profitable for institutions to serve low-income customers with small account balances, a segment previously considered unviable.
- Natural Language Processing (NLP) breaks down literacy and language barriers. NLP-powered chatbots and voice assistants allow users in rural or developing regions to interact with financial services using local dialects and natural speech, rather than complex written forms.

In conclusion, AI technologies hold the potential to expand access to financial services and empower individuals and communities to achieve greater financial security and prosperity. However, it is essential to ensure that AI solutions are developed and deployed in an ethical and responsible manner, with appropriate safeguards in place to protect consumer privacy and mitigate potential biases. By harnessing the power of AI, policymakers, financial institutions, and other

stakeholders can work towards building more inclusive and resilient financial systems that benefit everyone.

3. Embedded Finance, AI, and Real-Time Cross-Border Payments

The landscape of global banking is undergoing a paradigm shift from traditional, siloed banking interfaces to "Embedded Finance"—the seamless integration of financial services into non-financial platforms. This phenomenon is rapidly moving banking into the background of everyday life; consumers now book rides, order groceries, and manage freelance work through platforms that handle complex payment flows invisibly. According to recent industry analysis by Bain & Company, the embedded finance market is projected to facilitate over \$7 trillion in transaction value by 2026. However, as these platforms increasingly operate on a global scale, they face a critical infrastructure challenge: the reliance on real-time cross-border rails to settle transactions between workers, merchants, and consumers in different jurisdictions.

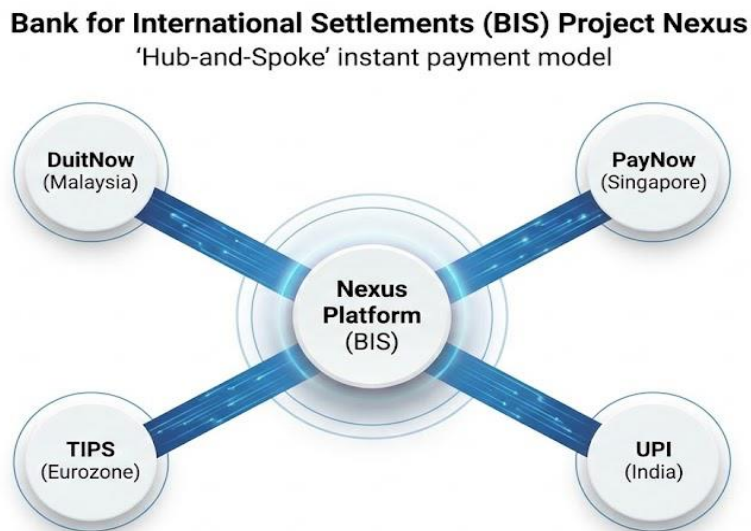
The expectation for the digital economy is immediacy. If a ride-hailing app pays a driver instantly in New York, a gig-worker in Manila expects the same speed from a US-based platform. Yet, the traditional cross-border payment architecture—reliant on a complex web of correspondent banks—often operates on T+2 settlement cycles with opaque fees. This latency creates a fundamental friction point for embedded finance. To bridge the gap between user expectations of instantaneity and the realities of international settlement, Artificial Intelligence (AI) has emerged as the critical orchestration layer. AI is no longer just an add-on; it is the central nervous system required to manage risk checks, routing decisions, and liquidity forecasting in milliseconds, ensuring that compliance and security do not become bottlenecks for the user experience.

3.1 AI in Real-Time Cross-Border Infrastructure and Interoperability:

To understand the necessity of AI, one must first examine the inefficiencies of the legacy model. The traditional correspondent banking system relies on *Nostro* and *Vostro* accounts—accounts that

banks hold with one another to facilitate settlement in foreign currencies. This system is capital inefficient, trapping billions of dollars in "pre-funded" liquidity, and operates largely on serial message processing. For an embedded finance platform, this means a payment might pass through three or four intermediary banks, each deducting a fee and adding a time delay.

The solution lies in linking domestic Instant Payment Systems (IPS) globally. **Project Nexus**, a flagship initiative by the Bank for International Settlements (BIS) Innovation Hub, exemplifies this architectural shift. Rather than building bilateral links between every country, Nexus offers a standardized "hub-and-spoke" connectivity model. This allows domestic systems—such as Malaysia's *DuitNow*, Singapore's *PayNow*, and eventually the Eurozone's *TIPS*—to interoperate seamlessly.



Picture 2

Within this high-speed architecture, AI serves two critical functions:

1. **Intelligent Routing and Liquidity Management:** AI algorithms analyze volatility in FX markets and liquidity positions across different currency corridors in real-time. They can predict the optimal route for a payment to minimize cost and latency, much like a GPS navigates traffic.

2. **ISO 20022 Data Enrichment:** The global shift to the ISO 20022 messaging standard provides richer, structured data for every transaction. Unlike older formats, which often truncated remittance information, ISO 20022 carries detailed data fields. AI models trained on this granular data can perform "pre-validation," predicting with high accuracy whether a payment will fail due to incorrect formatting or sanctions screening *before* it is sent. This drastically reduces the "failed payment" rate, which currently plagues roughly 5% of all cross-border transactions, ensuring the seamless experience embedded finance user's demand.

3.2 AI for Financial Integrity in Fragmented Markets:

While embedded finance accelerates capital flow, it introduces significant "speed risks" regarding financial crime. In a cross-border context, data silos allow criminals to exploit gaps between jurisdictions. A money laundering scheme might involve "smurfing"—breaking large sums into small, inconspicuous amounts—across accounts in three different countries. In the legacy system, a bank in Country A only sees the initial transfer, and a bank in Country B only sees the receipt, meaning neither spots the criminal pattern.

To counter this, AI utilizes **Graph Neural Networks (GNNs)** to analyze the payment network holistically rather than as isolated transfers. **Project Aurora**, another pivotal BIS initiative, demonstrated the power of this approach. Project Aurora explored how graph-based models could identify hidden money-laundering structures by treating accounts as "nodes" and payment flows as "links." By analyzing the topology of these connections, the AI can detect complex "mule networks" that traditional rule-based systems miss. In testing, these graph-based models identified approximately twice the number of potential laundering networks compared to standard methods.

However, sharing transaction data across borders triggers complex privacy regulations, such as the GDPR in Europe. To solve this, Project Aurora utilized **Privacy-Enhancing Technologies (PETs)**, specifically **Federated Learning**. This approach allows AI models to be trained locally within each institution's secure environment. Instead of sharing raw customer data (which remains private), the banks share only the updated model parameters (the mathematical learnings). The

central AI model aggregates these learnings to become smarter without ever "seeing" the underlying sensitive data. For embedded finance providers, this is a breakthrough: it allows them to plug into a global, AI-driven monitoring framework that preserves user privacy while robustly defending the integrity of the financial system.

3.3 Cross-Border Data Portability and "Credit Passports":

A major limitation in the current global embedded finance ecosystem is the lack of credit history portability. In a globalized economy, labor is mobile; millions of potential borrowers are migrants, international students, or cross-border freelancers. When these individuals move to a new country, they often face "thin file" status. Traditional credit bureaus in the host country cannot access their financial history from their home country, effectively resetting their creditworthiness to zero. This information asymmetry forces lenders to treat high-quality borrowers as high-risk, charging exorbitant interest rates or denying service entirely.

AI is bridging this gap through **Cross-Border Data Portability**, creating what are effectively "Credit Passports." Unlike the "alternative data" discussed in previous sections (which utilizes mobile behavior), this approach uses AI to translate and standardize *existing* formal financial data across borders.

Two prominent examples highlight this mechanism:

- **Mifundo (Europe):** In the European Union, cross-border lending has historically been low due to fragmented data. The fintech Mifundo is building a data layer that standardizes credit data across EU member states. Using AI to normalize different scoring methodologies (e.g., how a default is recorded in Germany vs. Estonia), it presents lenders with a unified view. They report that this transparency can reduce credit risk for banks by up to a factor of seven, unlocking capital for mobile citizens.
- **Nova Credit (Global):** On a global scale, Nova Credit has built integrations with major credit bureaus in countries like India, Mexico, and the UK. Their "Credit Passport" technology translates this foreign data into a format that US lenders (like American Express and JPMorgan Chase) can ingest directly into their underwriting models.

In these instances, AI acts as a **translation layer**, allowing creditworthiness to travel with the user. This capability is fundamental to the mature operation of global embedded finance platforms. It ensures that a software engineer moving from Bangalore to San Francisco can access a credit card or apartment lease immediately upon arrival, integrating them into the financial system of their new home instantly. By enabling this data portability, AI does not just improve banking efficiency; it supports global labor mobility and economic integration.

Thus, The intersection of embedded finance and AI is reshaping the infrastructure of global commerce. By replacing the rigid, slow, and siloed legacy systems with agile, AI-driven interoperability, the financial sector is moving toward a future where payments and credit are borderless. From the liquidity management of Project Nexus to the privacy-preserving fraud detection of Project Aurora and the credit mobility of Nova Credit, AI is the engine making financial globalization a practical reality for the individual user.

4. Risk of AI on Global Financial Institutions

Artificial Intelligence has become widely regarded as a phenomenon in the financial world causing impactful transformations for bankers and causing shockwaves in its ability to transform global financial markets. Artificial Intelligence is being utilized in major operations, user interactions and the assessments of danger and risks for a wide degree of people and companies including fintech's and bankers(SUERF Policy Note 2025). These technologies are implemented completely on the premise that they would improve efficiency to a large degree causing advances in many areas like in the AI's ability to think itself and the wide array of new services it could provide.(World Economic Forum 2025) Looking deeper, it's interesting to see the large level of dependence that giant financial institutions across the globe have on this and the way that the AI reshapes the risk landscape for these financial institutions. This section critically examines the various risks (systemic, operational, and governance) that AI imposes on the global financial ecosystem using large banks of data, research and findings. (Bank of England 2025)

4.1 The Systemic Risks stemming from AI

Artificial Intelligence is increasingly recognized as a transformative force in global markets, fundamentally altering operational efficiency, user interaction, and risk assessment architectures. These technologies are integrated on the premise of optimizing decision-making and expanding service capabilities. However, this deep integration creates a paradox: as financial institutions increasingly rely on autonomous algorithmic systems, the nature of systemic risk evolves from traditional liquidity concerns to complex, interconnected technological vulnerabilities.



Picture 3

An example which could be taken to represent the systemic risks of Artificial Intelligence in this field is if an algorithmic trading platform powered by artificial intelligence failed or malfunctioned leading to sudden liquidity shocks and stock crashes which could swiftly spread globally due to the ease and efficiency of connection worldwide. (LSE Working, Paper 2021)

High Frequency trading in the financial world powered by AI can lead to unexpected swings in the prices and cause distortions in stock, housing and overall asset pricing causing large alterations to the market stability which may have been previously present before the onset of Artificial Intelligence (LSE Working Paper, 2021). This shows a way in which Artificial Intelligence can lead to volatility in the market. Other ways that this poses a risk is where we start to see Artificial

Intelligence being involved in other parts of the financial market including fraud detection models and credit scoring. This wide range of fields increases the risk of herding where Artificial Intelligence is so largely involved in multiple industries, and one attack could cause a dramatic shock and various issues to the economy. This is where we can determine the fact that even though the level of connectedness between various fields because of Artificial Intelligence is useful it increases the risk that problems in a specific industry could cause a domino effect into multiple/the whole industry and lead to a global crisis(Bank Of England 2025).

4.2 The Data and Algorithmic Risks from Artificial Intelligence

The second critical vector of vulnerability arises from Data and Algorithmic Risk. This encompasses the hazards associated with dataset integrity, algorithmic processing, and privacy preservation. AI models are not objective; they are dependent on the quality of their training data. If historical biases are present in the source data, the resulting algorithms may perpetuate discriminatory outcomes—such as loan denial or stratified insurance pricing—thereby undermining regulatory standards for fair lending.

In today's society a major focus of organisations must be diverse and high quality data that represents a range of backgrounds and these models must be continuously monitored in case of negative outcomes(SUERF Policy Note 2025). Data privacy can also show a wide range of major challenges. This centralisation of customer, transaction and behavioural data in these artificial intelligence systems cause large risks in terms of cybercriminals. These are also very high value targets for criminals to attack. This is where cracks in the system like third party openings, poor encryption and lapses in internal controls can cause significant intrusions into a financial system and essentially even lead to a complete loss of public trust in the financial system. Researchers delve deeper into the risks where people may seek to corrupt the inputs and sources of data in a decision-making process which could lead to undermining of regulation and unapproved transactions. (Bank of England, 2025)

A prominent example occurred with the **Apple Card in 2019**, where the credit limit algorithm allegedly offered significantly lower limits to women compared to men with similar credit profiles, including the wife of Apple co-founder Steve Wozniak, despite them sharing assets. This

highlights the risk of 'proxy discrimination' where algorithms find correlations that humans might miss.

4.3 Opacity and Explainability Risks

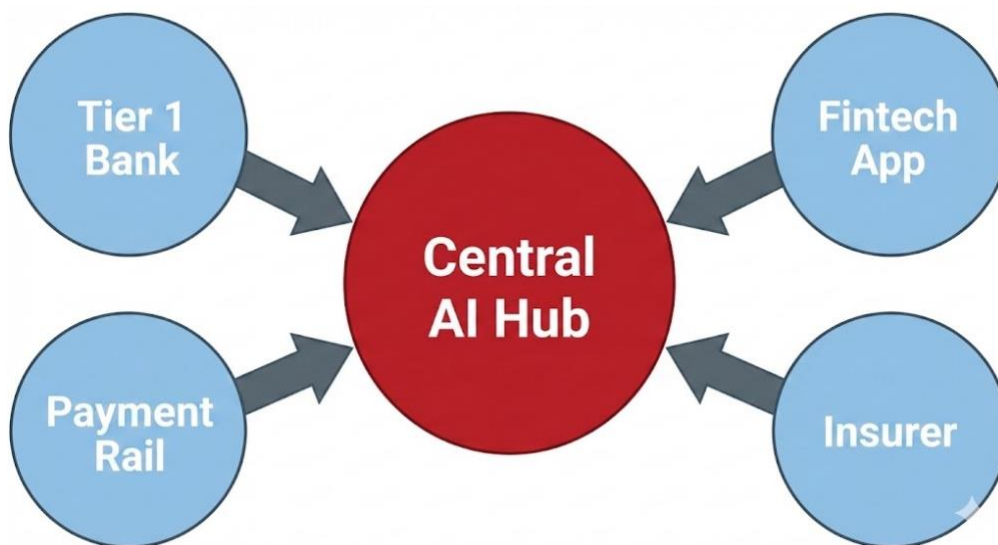
A critical challenge in AI adoption is 'Black Box' opacity—the difficulty humans face in interpreting the internal decision-making processes of deep learning models. This section examines the 'Black Box' phenomenon, where human operators struggle to interpret the internal decision-making processes of AI and then the risk that there isn't the ability for an understandable reason to be made after a decision is taken (Alan Turing Institute 2024, World Economic Forum 2025). When we investigate the AI models of the present and look at those with generating or deep learning techniques, we observe how complex outputs and decisions are made that are tough for humans to be able to understand. The opacity that arises because of this makes it very hard for both external regulators and internal regulators to understand how decisions can be made. When the logic of a model provided by an Artificial Intelligence can't be explained it leads to financial institutions in the global financial market failing to notice errors and risks that might be emerging causing firms to not be able to respond in time. This lack of explaining can lower market confidence and the ability to respond swiftly to untoward conditions (Financial Stability Board 2024). The Bank of England chooses to keep in mind that the use of smart, complex artificial intelligence systems bring certain challenges for the transparency and oversight causing the allowance of models to potentially allow collusion, manipulation and even forms of market abuse (Bank of England 2025). Financial institutions are asked to have proper documentation on the architecture of their AI models to ensure the transparency of reporting of decision making and to help develop tools to aid human decision makers and external regulators in understanding models (Alan Turing Institute 2024, SUERF Policy Note 2025).

4.4 Cybersecurity and Concentration Risks

When institutions have increased dependence on a smaller pool of important Artificial Intelligence platforms and various shared platforms there is an increased risk of market concentration (SUEF Policy Note 2025, Financial Stability Board 2024). The vulnerabilities in the systems of a company or in critical algorithms could put whole financial markets at risk (World Economic Forum 2025). The concentration of this not only causes risk to the resilience and protection during technical

incidents but may also lead to competition and innovation being stifled. The threats to Artificial Intelligence or model supply chains lead to huge challenges arising. Attacks on the network may be very likely having a chance of being coordinated to destroy whole networks and the financial system (EFR White Paper 2025).

This is largely shown in studies done internationally by acclaimed institutions where there may be coordinated attacks leading to compromising of credit scoring systems and various high level financial processes like loan eligibility and more (Financial Stability Board 2024, LSE Working Paper 2021). These attacks on trading models can cause rapid shocks and changes in the market and may cause rapid selling off and asset freezes across multiple financial markets or the financial market (LSE Working Paper, 2021). This continues to back up the need for investment in proper cybersecurity in this industry as well sophisticated levels of high level, multi-layered risk management and it's also clear that financial institutions should ensure that these systems are regularly tested for issues to understand work points (EFR White Paper 2025, Bank of England 2025).



Picture 4

According to 2024 industry reports, financial firms faced a **25% surge** in advanced cyberattacks compared to the previous year. Furthermore, reports indicate that nearly **97% of the largest U.S. banks** suffered breaches linked to third-party vendors, validating the fear of concentrated supplier risk.

4.5 The Risks in Regulation, Ethics and Governance stemming from AI

The ultimate category focuses on these three risks. Regulatory risks are the risks where changes in regulation may affect a business negatively as a result of AI. Ethics risks are the risk of harm to an organization because of a violation in ethical values or standards (Bank of England 2025, SUEF Policy Note 2025). Governance risks refer to the negative consequences that may occur due to poor internal systems and procedures that govern the management and direction of an organization. The implementation of AI in finance has largely been very swift, almost too swift causing oversight in existing frameworks and gaps in the accountability and documentation of the various processes taking place. Internal financial institutions and external regulators may have a hard time keeping up with evolving models further increasing by the new force of generative AI and reinforcement learning approaches. This is where the SUEF policy brief comes into play providing a proper framework including recommendations such as:

1. Assigning clear roles and responsibilities for AI monitoring in the organization;
2. Ensuring regular audits and external review of Artificial Intelligence Systems.
3. Providing channels for customers to fight against AI made decisions
4. Coordinating internationally to ensure this doesn't become an issue.

There is also a request for further research, where it has been called for regulators to go towards traditional policy objectives, causing new mandates around data privacy, algorithmic bias and the various economic impacts of AI driven automation. It is largely recommended that these policy makers test their systems for various scenarios (SUEF Policy Note, 2025).

Thus, artificial Intelligence showcases a range of great opportunities for financial institutions across the world improving overall efficiency and market access across the sector. There are risks and disadvantages to this which may be extremely dangerous if left unattended for long periods of time. These risks are systemic, data, algorithmic, cyber related and governance. This is where it proves there must be a collaborative effort from external regulators and internal financial institutions to gain benefits from Artificial Intelligence. This includes the forming of new, flexible frameworks for improving AI systems. Only by this collaboration and integration would there be

no undermining of global financial stability. (SUERF Policy Note 2025, Financial Stability Board 2024)

5. Main Results

Based on the qualitative synthesis of industry reports and case studies, this research identifies six primary findings regarding the integration of AI in global finance:

- **AI Dramatically Enhances Fraud Detection:** Machine learning and network-based models (such as Graph Neural Networks) detect complex fraud patterns in real-time. These systems significantly outperform traditional rule-based methods, strengthening financial integrity against sophisticated money laundering networks.
- **Algorithmic Scoring Expands Financial Inclusion:** By analyzing alternative data (mobile usage, utility payments) and cash-flow behavior, AI enables accurate credit assessments for the "unbanked" and migrant populations. This approach, exemplified by firms like Tala, reduces default risks while democratizing access to capital.
- **Digital Banking is the Gateway to Access:** With over 3.6 billion users globally, digital banking has transitioned from a convenience to a necessity. The shift to API-first and cloud-based platforms has reduced operational costs and widened financial access to previously underserved regions.
- **Real-Time Cross-Border Payments are Scalable but Fragile:** Initiatives like the BIS Project Nexus demonstrate that AI can enable scalable interoperability between national payment systems. However, the requirement for millisecond-level screening introduces new operational vulnerabilities and cyber risks.
- **Emergence of New Systemic Risks:** The sector's heavy reliance on a concentrated group of shared AI models and cloud providers increases "correlation risk." This creates the potential for **algorithmic herding**, where simultaneous model failures could destabilize the broader market.
- **The "Black Box" Governance Challenge:** Data quality issues and model opacity remain critical barriers. AI systems risk perpetuating historical biases, and their complexity limits

auditability. This necessitates a shift toward "macro-prudential" regulation that focuses on transparency and explainability.

6. Conclusion

The global financial system is undergoing a structural shift in which artificial intelligence has moved from the periphery of operations to the core of how trust, risk, and creditworthiness are managed across borders. AI-powered fraud detection and algorithmic credit scoring now underpin the growth of digital banking, embedded finance, and real-time cross-border payment rails. At the micro level, these systems deliver tangible gains: they enable millisecond-level screening on high-velocity networks, detect complex fraud patterns that human and rules-based systems routinely miss, and open new avenues for financial inclusion by extending credit and payment access to customers, merchants, and regions historically ignored by traditional banking models.

Yet these same technologies also reconfigure systemic risk in ways that existing prudential frameworks are not fully prepared to address. Embedded finance blurs the boundaries between banks, fintechs, big techs, and non-financial platforms, creating a dense ecosystem of interconnected services that rely on shared data, common AI tooling, and concentrated cloud and infrastructure providers. As institutions converge on similar models, datasets, and vendors, the financial system drifts toward algorithmic monoculture: correlated model errors, synchronized risk responses, and large-scale algorithmic herding become not just possible, but likely in periods of stress. In such a landscape, what appears as micro-prudential improvement—more accurate fraud filters, sharper risk scores—can paradoxically increase macro-systemic fragility.

Data quality, embedded bias, and model opacity further deepen this tension. If the data that fuel AI systems mirror historical exclusion or geopolitical imbalance, and if the resulting models operate as opaque black boxes, the outcomes are both unfair and potentially destabilizing. Bias in algorithmic credit scoring can scale into structural exclusion of particular regions, demographics, or cross-border flows, while opaque fraud models can silently reshape transaction patterns and liquidity routes. Combined with real-time payment architectures that leave little room for manual oversight, these properties create tightly coupled, high-speed systems in which small design flaws or shocks can propagate quickly across institutions and jurisdictions.

The central challenge, therefore, is not whether AI should be integrated into global finance, but how to govern this integration so that its stabilizing features outweigh its destabilizing potential. Meeting this challenge requires a distinctly macro-prudential approach to AI. Regulators and policymakers must look beyond individual model performance to focus on model convergence, infrastructure concentration, and cross-border dependencies. They will need to promote diversity in modeling approaches and data sources, mandate robust transparency and auditability standards, and design stress tests that explicitly account for AI-driven behavioral dynamics under stress scenarios. At the infrastructure level, resilience demands redundancy in cloud and service providers, well-defined failover strategies, and coordinated incident response mechanisms spanning borders and institutional types.

If these governance, technical, and regulatory safeguards are put in place, AI can function not as an opaque and fragile backbone for embedded finance, but as a disciplined, adaptive infrastructure that strengthens the integrity of cross-border financial flows. The future of AI-driven digital banking and embedded finance will be decided not only by computational advances, but by whether global institutions choose to design for diversity, transparency, and resilience rather than speed and efficiency alone. Done well, AI can anchor a more inclusive, efficient, and robust global financial system; done poorly, it risks becoming a tightly coupled, algorithmic substrate through which localized errors and biases scale into systemic crises. The task ahead is to deliberately choose the former path.

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Figure 1: *Leveraging AI for Credit Scoring (Source: Raza & Tariq, 2022)*

Figure 2: *Conceptual Diagram of the BIS Project Nexus 'Hub-and-Spoke' Model. (Generated by Google Gemini, 2025).*

Figure 3: *Systemic risks of Artificial Intelligence. (Generated by Google Gemini, 2025)*

Figure 4: *Concentration risks. (Generated by Google Gemini, 2025)*

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